Lab 3.1 source pack

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

system-description.txt

SYS-01 revision 1 | fictional chemical dosing skid
Function: deliver a controlled dose to a training water line. Elements: tank T-1; metering pump DP-1; check valve CV-1; flow sensor FT-1. Current controls: tank level alarm checked each shift; weekly pump check; monthly valve inspection; annual sensor calibration. Control effectiveness is unknown. Failure candidates: no flow, excessive flow, reverse flow, sensor drift. Fluid identity, real exposure limits and approved materials are outside the case. Treat all consequences and ratings as workshop hypotheses requiring engineering review.

fmea-template.csv

id: F01; function: Deliver controlled dose; failure_mode: No flow; effect: Underdose; cause: Pump failure; current_control: Weekly check; S: 8; O: 4; D: 6; RPN: ; action: Review redundancy and alarm evidence; owner: Engineer A; due_date: 2026-09-12; evidence: SYS-01 rev1; status: Proposed

id: F02; function: Prevent reverse flow; failure_mode: Reverse flow; effect: Cross contamination; cause: Valve stuck; current_control: Monthly inspection; S: 9; O: 3; D: 7; RPN: ; action: Review backflow protection; owner: Engineer B; due_date: 2026-09-12; evidence: SYS-01 rev1; status: Proposed

id: F03; function: Measure dose; failure_mode: Sensor drift; effect: Incorrect adjustment; cause: Calibration drift; current_control: Annual calibration; S: 7; O: 5; D: 5; RPN: ; action: Review calibration frequency; owner: Engineer C; due_date: 2026-09-15; evidence: SYS-01 rev1; status: Proposed

rating-scales.txt

IES TRAINING FMEA SCALE revision 1
This is a simplified classroom scale, not an industry or company standard. Use integer ratings 1-10.
Severity: 1 negligible training impact; 4 local rework; 7 major functional loss; 10 potentially catastrophic consequence. Intermediate ratings need written justification.
Occurrence: 1 remote; 4 occasional; 7 frequent; 10 persistent. These are qualitative classroom anchors, not probabilities.
Detection: 1 almost certain detection before the effect; 4 likely; 7 unlikely; 10 no effective detection. A larger D means poorer detectability.
RPN = S x O x D. Review high-severity scenarios separately even when RPN is lower. Proposed actions do not reduce ratings until their effectiveness has been verified.